Market Risk Analysis

Project Report

Project Details:

Project Name	Market Risk Analysis Project
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1. Project Background

Investors face significant challenges due to market risk, a form of risk that stems from the inherent volatility in asset prices. This volatility is influenced by a wide range of factors, including economic events such as changes in interest rates, inflation, and economic growth, geopolitical developments such as political instability, international conflicts, or policy changes, and shifts in investor sentiment including market psychology, news, and social trends. These factors can lead to rapid, unpredictable fluctuations in asset values, which can directly impact the financial performance of an investment portfolio.

The inability to accurately predict and manage these market fluctuations can result in substantial losses for investors, particularly in times of economic uncertainty or crisis. As a result, it becomes critical for investors to not only recognize the presence of market risk but also to develop strategies for mitigating its impact.

To optimize investment strategies, investors need to possess a robust understanding of market risk dynamics. This involves analysing historical data, market trends, and underlying economic indicators to anticipate potential risks and prepare for adverse scenarios.

By properly analysing these factors and incorporating risk management techniques, investors can make more informed decisions, create more resilient portfolios, and ultimately, enhance long-term financial performance while minimizing exposure to potentially catastrophic losses.

2. Objective

The goal of this analysis is to perform a Market Risk Assessment on a portfolio of Indian stocks using Python. By utilizing historical stock price data, the analysis aims to assess market volatility and risk exposure. Statistical measures such as mean and standard deviation will be employed to provide a clearer picture of the performance of individual stocks and the overall portfolio's variability.

Through this analysis, investors can pursue the following objectives:

- **Risk Assessment:** Evaluate the historical volatility of individual stocks and the entire portfolio.

- **Portfolio Optimization:** Leverage insights from the Market Risk Analysis to improve risk-adjusted returns.

- **Performance Evaluation:** Assess the effectiveness of portfolio management strategies in managing market risk.

- **Portfolio Performance Monitoring:** Continuously track portfolio performance and adjust in response to evolving market conditions and changing risk preferences.

By achieving these objectives, investors can make more informed decisions, optimize their portfolios, and better manage risk to enhance long-term returns.

3. Data description

- The dataset contains weekly stock price data for 5 Indian stocks over an 8-year period.

- The dataset enables us to analyse the historical performance of individual stocks and the overall market dynamics.

4. Data structure

After importing NumPy and Pandas libraries for tasks like data reading, data computation and data manipulation and loading data, the following observations regarding the data structure were noted.

	Date	ITC_Limited	Bharti_Airtel	Tata_Motors	DLF_Limited	Yes_Bank
0	28-03-2016	217	316	386	114	173
1	04-04-2016	218	302	386	121	171
2	11-04-2016	215	308	374	120	171
3	18-04-2016	223	320	408	122	172
4	25-04-2016	214	319	418	122	175

Table 1. Data structure

OBSERVATION	VARIABLES
418	6

5. Data types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Date           418 non-null    object
 1   ITC_Limited    418 non-null    int64
 2   Bharti_Airtel  418 non-null    int64
 3   Tata_Motors    418 non-null    int64
 4   DLF_Limited    418 non-null    int64
 5   Yes_Bank       418 non-null    int64
dtypes: int64(5), object(1)
memory usage: 19.7+ KB
```

Table 2. Data types

- The data set contains 6 variables indexed from 0-5.
- The date column is in object format and must be converted into data format.
- All the other columns are numerical columns of integer data type.
- The dataset does not contain any missing values.

6. Statistical Summary

	ITC_Limited	Bharti_Airtel	Tata_Motors	DLF_Limited	Yes_Bank
count	418.000000	418.000000	418.000000	418.000000	418.000000
mean	278.964115	528.260766	368.617225	276.827751	124.442584
std	75.114405	226.507879	182.024419	156.280781	130.090884
min	156.000000	261.000000	65.000000	110.000000	11.000000
25%	224.250000	334.000000	186.000000	166.250000	16.000000
50%	265.500000	478.000000	399.500000	213.000000	30.000000
75%	304.000000	706.750000	466.000000	360.500000	249.750000
max	493.000000	1236.000000	1035.000000	928.000000	397.000000

Table 3. Statistical Summary

The following observations were made:

- **ITC_Limited**: Over the past 8 years, the stock price ranged from a low of 156 to a high of 493, with an average of 278. It exhibits a relatively low standard deviation of 75, indicating less volatility in its price movement.

- **Bharti_Airtel**: The stock price fluctuated between a minimum of 261 and a maximum of 1236, with an average of 528. The standard deviation of 226 suggests higher volatility compared to ITC, reflecting more significant price swings over time.

- **Tata_Motors**: Stock prices for Tata Motors ranged from 65 at the lowest to 1035 at the highest, with an average of 368. With a standard deviation of 182, the stock shows moderate volatility, though not as much as Bharti Airtel.

- **DLF_Limited**: DLF's stock prices saw a minimum of 110 and a maximum of 928, with an average of 276. The standard deviation of 156 indicates moderate volatility, suggesting price fluctuations but with more stability than some other stocks.

- **Yes_Bank**: Yes Bank experienced a wide range in stock prices, from a low of 11 to a high of 397, with an average price of 124. The stock has a relatively high standard deviation of 130, highlighting significant volatility, which may indicate a riskier investment.

7. Stock Price Graph Analysis

7.1. ITC Limited

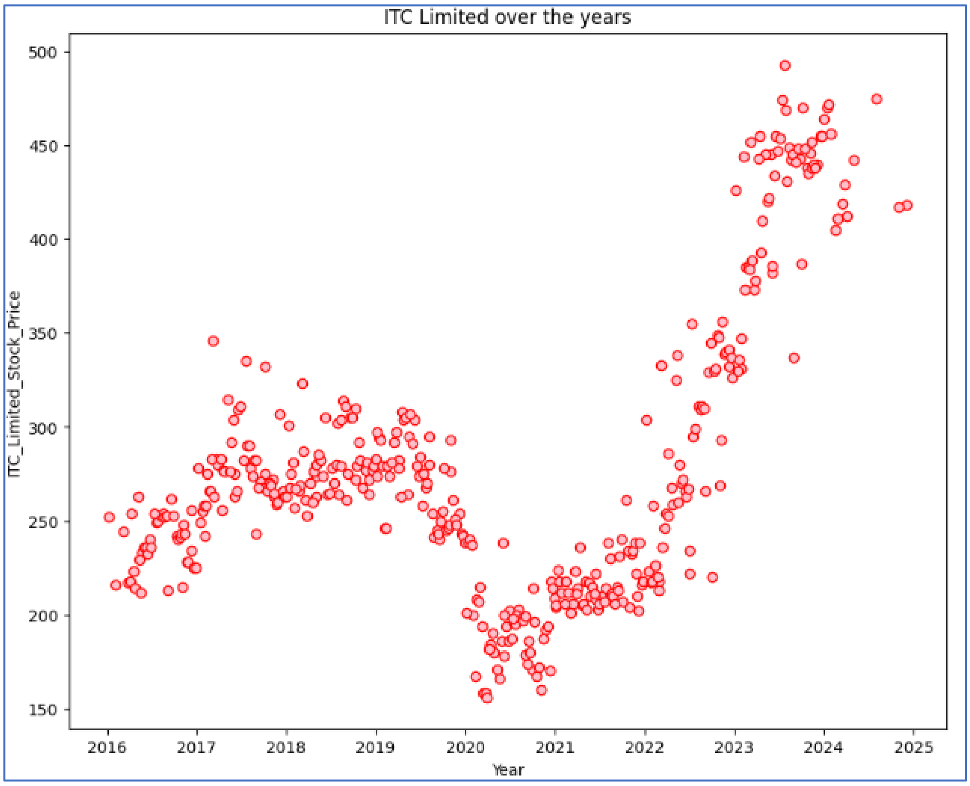

Fig 1. Stock Price vs Time: ITC Limited

OBSERVATIONS:

- The ITC Limited stock prices were relatively low, around 200, in 2016.
- From 2017 to 2019, there was a steady increase. However, in 2020-2021, the prices dropped to 150, likely due to the impact of the pandemic.
- Starting in 2022, the prices began to rise gradually, reaching 500, with a positive upward trend continuing into 2025.
- The overall upward movement since 2017, despite the temporary setback, indicates a long-term positive trajectory, with the stock gradually overcoming past challenges.
- The ongoing positive trend toward 2025 indicates sustained market optimism, likely driven by improved fundamentals, economic recovery, or positive investor sentiment.

7.2. Bharti Airtel

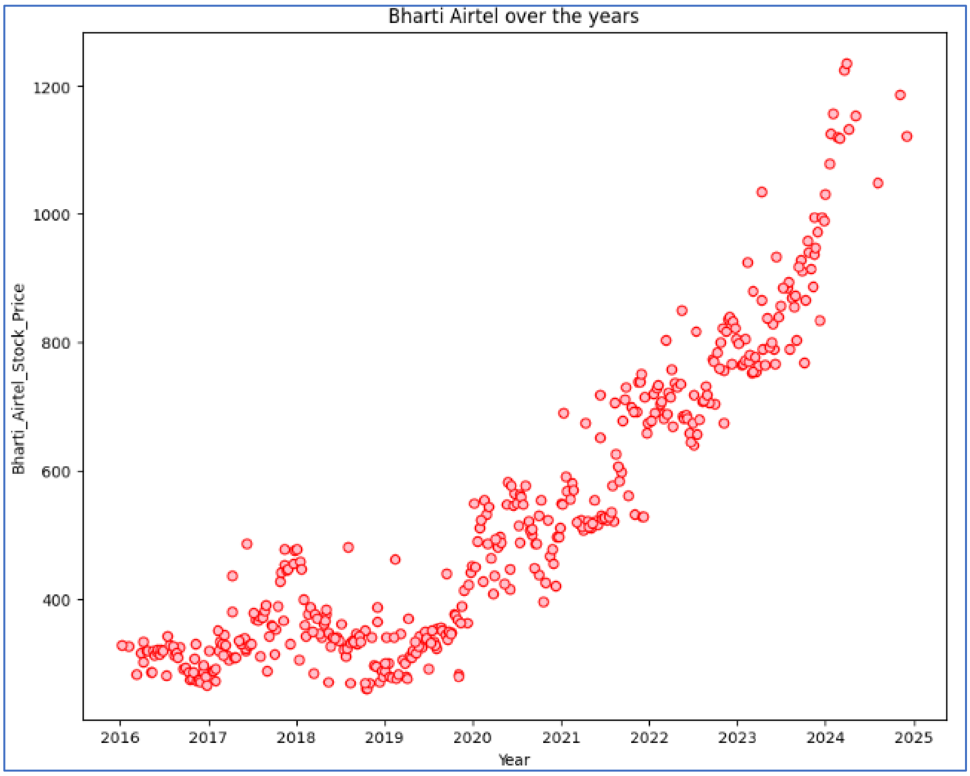

Fig 2. Stock Price vs Time: Bharti Airtel

OBSERVATIONS:

- The Bharti Airtel stock prices remained stable between 300 and 400 from 2016 to 2019. This period reflects a phase of stability which indicates a lack of major market catalysts or significant growth opportunities during this time.
- Starting in 2020, the prices began to rise steadily, with a consistent linear increase. The significant price increase over the five years implies that the market has gained confidence in the stock's long-term potential, with consistent growth building investor trust in its prospects.
- In 2020, the stock was priced around 400, and by 2025, it had climbed to 1200. The linear nature of the growth could indicate stable, predictable performance rather than sudden volatility.

7.4. Tata Motors

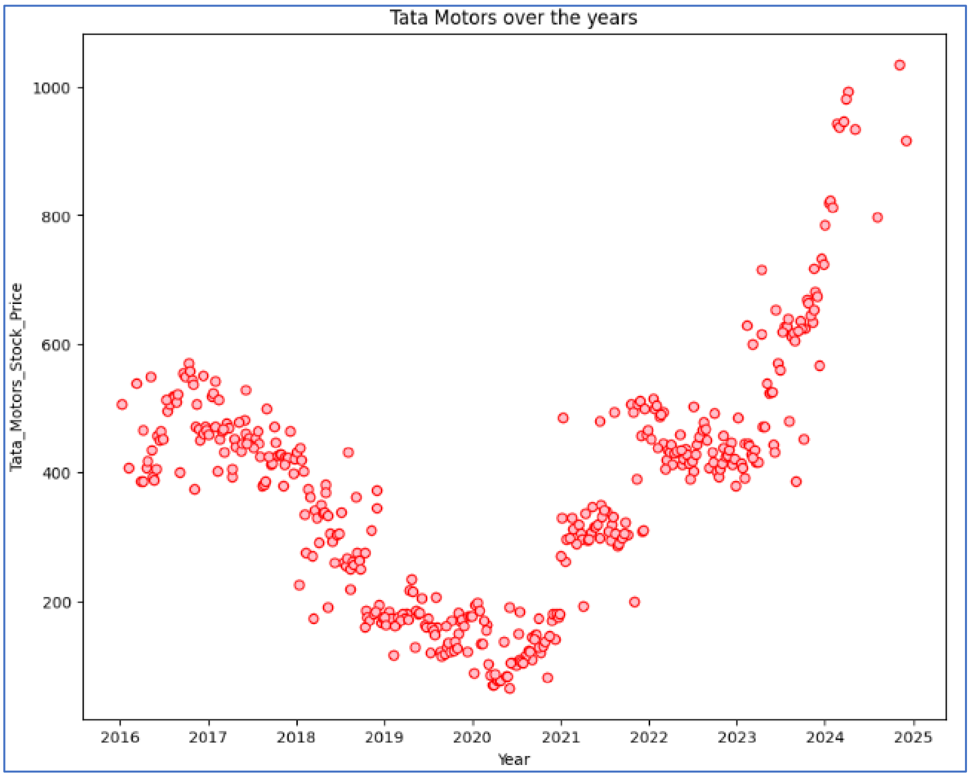

Fig 3. Stock Price vs Time: Tata Motors

OBSERVATIONS:

- From 2016 to 2020, the stock experienced a decline, with prices dropping from 400-500 to around 100 in 2020-2021, likely due to the pandemic's impact on Tata Motors, likely due to supply chain disruptions, lower demand, or broader economic challenges faced by the automotive sector.
- However, there has been a gradual recovery from 2021 to 2025, with the price rising from 200 to 1000. The steady rise indicates a strong recovery, potentially driven by improved market conditions, the company's adaptation to post-pandemic challenges, and a rebound in consumer demand or production.
- The gradual increase suggests that the company is on a path of sustained growth, possibly benefiting from innovation, expanding markets, or improving financial health, with a positive outlook for the next few years.

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7.5. DLF Limited

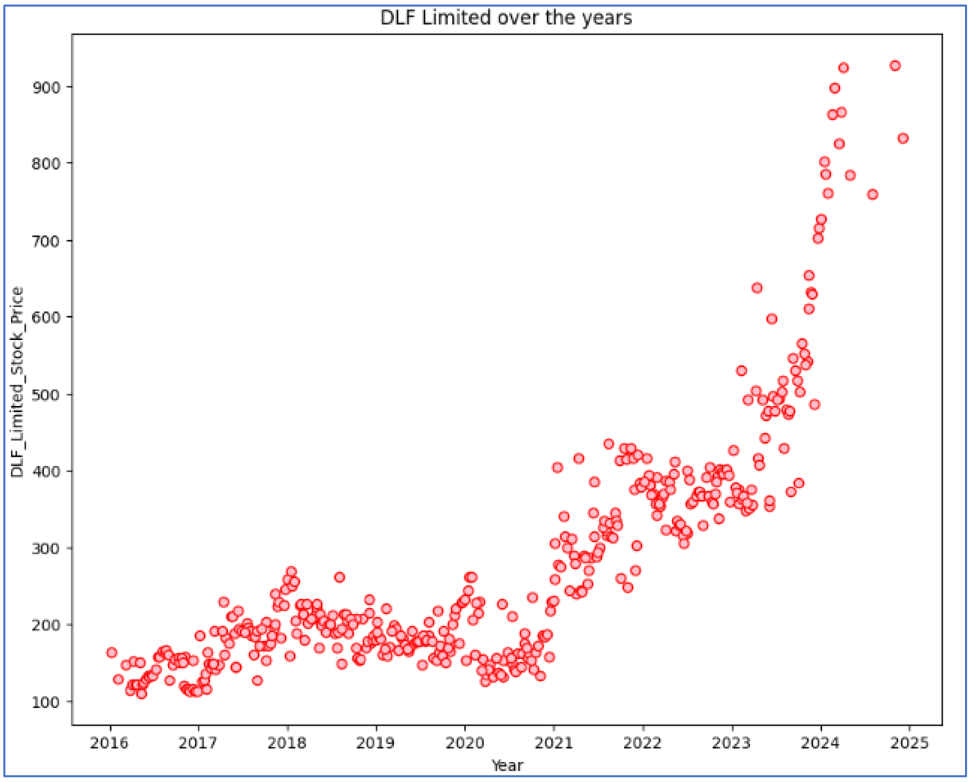

Fig 4. Stock Price vs Time: DLF Limited

OBSERVATIONS:

- The stock prices of DLF Limited remained relatively low, ranging between 100 and 200 until 2021, reflecting a steady or almost flat movement suggesting a lack of significant market drivers or major company developments during this time.
- In 2021, the price was around 150, and it increased to 400 by 2022, before dropping to 300 in 2023. The drop indicates some volatility, which could be attributed to market corrections, external factors, or temporary setbacks in the company's operations.
- However, the stock showed a steady rise from 2023 to 2025, reaching 900 by the end of the period. The subsequent rise from 300 to 900 indicates a strong recovery and investor confidence, likely driven by favourable market conditions, strategic initiatives, or positive news related to the company's performance.

7.6. Yes Bank

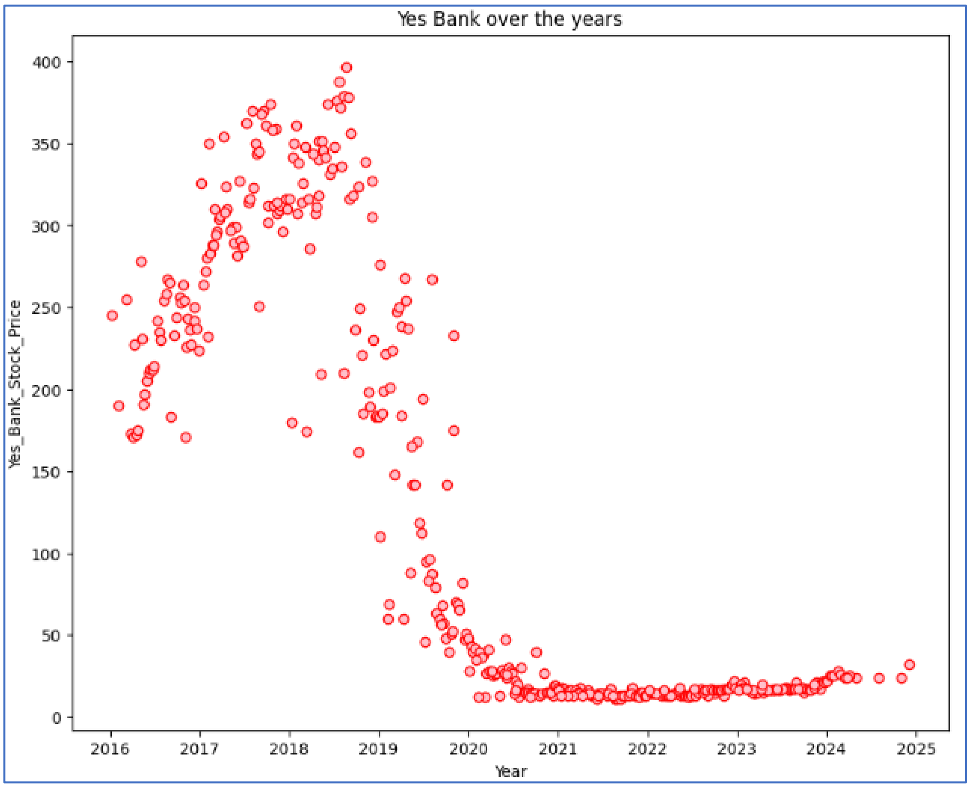

Fig 5. Stock Price vs Time: Yes Bank

OBSERVATIONS:

- The Yes Bank's stock prices gradually increased from 150 to 400 between 2016 and 2019. This period indicates a period of optimism, possibly driven by improvements in the bank's performance or market sentiment towards the financial sector.
- However, from 2019 to 2021, there was a steady decline, with prices falling below 50. This suggests a major setback, likely due to financial challenges, regulatory issues, or a loss of investor confidence, which could have been exacerbated by external factors like the pandemic.
- From 2020 to 2025, the stock has remained stagnant, showing little to no increase, and has consistently stayed below the 50 marks. This indicates that Yes Bank has struggled to recover or regain market trust leaving the stock largely undervalued or stagnant during this period.

8. Stock Returns Calculation and Analysis

Return refers to the gain or loss made on an investment over a specific period, expressed as a percentage of the initial investment. It is a key metric for investors to assess the profitability of an asset or portfolio.

8.1. Calculating Logarithmic Return

Logarithmic Return, also known as log return, is used for measuring the return on a stock, using the natural logarithm of the ratio of the stock's ending price to its beginning price.

By converting the date from object format to date format, we examine the data structure, which consists of 418 rows and 5 columns.

The logarithmic return is calculated for all the stocks which is as shown below.

	ITC_Limited	Bharti_Airtel	Tata_Motors	DLF_Limited	Yes_Bank
0	NaN	NaN	NaN	NaN	NaN
1	0.004598	-0.045315	0.000000	0.059592	-0.011628
2	-0.013857	0.019673	-0.031582	-0.008299	0.000000
3	0.036534	0.038221	0.087011	0.016529	0.005831
4	-0.041196	-0.003130	0.024214	0.000000	0.017291
5	0.009302	0.024769	-0.024214	0.055791	0.082238
6	-0.013986	-0.006135	-0.019803	-0.015625	-0.037538
7	-0.004706	-0.015504	-0.015114	-0.040166	0.042787
8	0.094452	-0.025318	-0.012772	0.016261	0.030930
9	0.012793	0.019048	0.040308	0.039531	0.039806

Table 4. Logarithmic Return

OBSERVATIONS:

- The logarithmic return calculation shows positive and negative values which indicates the direction of the change in the stock's price:
- A positive logarithmic return means the stock's price has increased from the previous period. It signifies a gain or an upward movement in the stock's value.
- A negative logarithmic return means the stock's price has decreased from the previous period. It signifies a loss or a downward movement in the stock's value.

8.2. Calculating Stock Means

Stock means is the average price of a stock over a specific period.

The stock means is calculated for all the stocks which is as shown below.

```
ITC_Limited       0.001634
Bharti_Airtel     0.003271
Tata_Motors       0.002234
DLF_Limited       0.004863
Yes_Bank         -0.004737
dtype: float64
```

Table 5. Stock Means

OBSERVATIONS:

- **ITC Limited:** The average weekly return of ITC Limited is 0.001634 which means on average, the stock's value increases by 0.001634 per week.
- **Bharti Airtel:** The average weekly return of Bharti Airtel is 0.003271 which means the stock's value increases by 0.003271 on average per week.
- **Tata Motors:** The average weekly return of Tata Motors is 0.002234. The stock increases by 0. 002234 on average each week.
- **DLF Limited:** The average weekly return of DLF Limited is 0.004863. This means that, on average, the stock increases by 0.004863 per week.
- **Yes Bank:** The average daily return of Yes Bank is -0.004737. This negative value indicates that, on average, Yes Bank's stock price decreases by 0.004737 per week over the period.

This parameter helps investors to assess whether a stock is trending positively or negatively over the period under analysis. Positive mean returns indicate growth, while negative mean returns suggest decline.

8.3. Calculating Stock Standard Deviation

Stock Standard Deviation quantifies the volatility or risk of a stock's price over a specific period. It indicates how much the stock's price deviates from its mean price.

A higher standard deviation means the stock's price is more volatile, meaning it fluctuates more widely from its average value. A lower standard deviation means the stock's price is less volatile and tends to stay closer to the mean.

The stock standard deviation is calculated for all the stocks which is as shown below.

```
ITC_Limited       0.035904
Bharti_Airtel     0.038728
Tata_Motors       0.060484
DLF_Limited       0.057785
Yes_Bank          0.093879
dtype: float64
```

Table 6. Stock Standard Deviation

OBSERVATIONS:

- **ITC Limited:** The stock price typically fluctuates by 0.035904 from its average return over the period considered. It has relatively lower volatility, suggesting that it is a more stable investment compared to other stocks.
- **Bharti Airtel:** The standard deviation is 0.038728, which shows a slightly higher volatility compared to ITC. This indicates moderate risk but still relatively stable.
- **Tata Motors:** They have a standard deviation of 0.060484 indicating higher volatility. This suggests that Tata Motors has more risk, and investors might see larger price swings in either direction.
- **DLF Limited:** The standard deviation is 0.057785 showing moderate-to-high volatility. This implies a higher level of risk than ITC and Bharti Airtel but less than Tata Motors.
- **Yes Bank:** The standard deviation is 0.093879 which is highest among the stocks. This indicates the highest level of volatility and risk.

ITC Limited and Bharti Airtel experience smaller price fluctuations and are less risky. Yes Bank has the highest standard deviation, indicating that its stock price is highly volatile and subject to larger swings, which increases the risk for investors. Tata Motors and DLF Limited have moderate levels of volatility, meaning that their stock prices experience moderate fluctuations from their average prices.

Below is the data frame comprising Stock means and Stock standard deviation.

	Average	Volatility
ITC_Limited	0.001634	0.035904
Bharti_Airtel	0.003271	0.038728
Tata_Motors	0.002234	0.060484
DLF_Limited	0.004863	0.057785
Yes_Bank	-0.004737	0.093879

Table 7. Stock Means & Stock Standard Deviation

8.4. Mean vs Standard Deviation Plot

Mean vs standard deviation plot analysis can provide insights into how each stock behaves relative to its return potential and risk level. This information helps to choose stocks that align with risk tolerance and investment goals.

Below is the mean vs standard deviation plot for all stock returns.

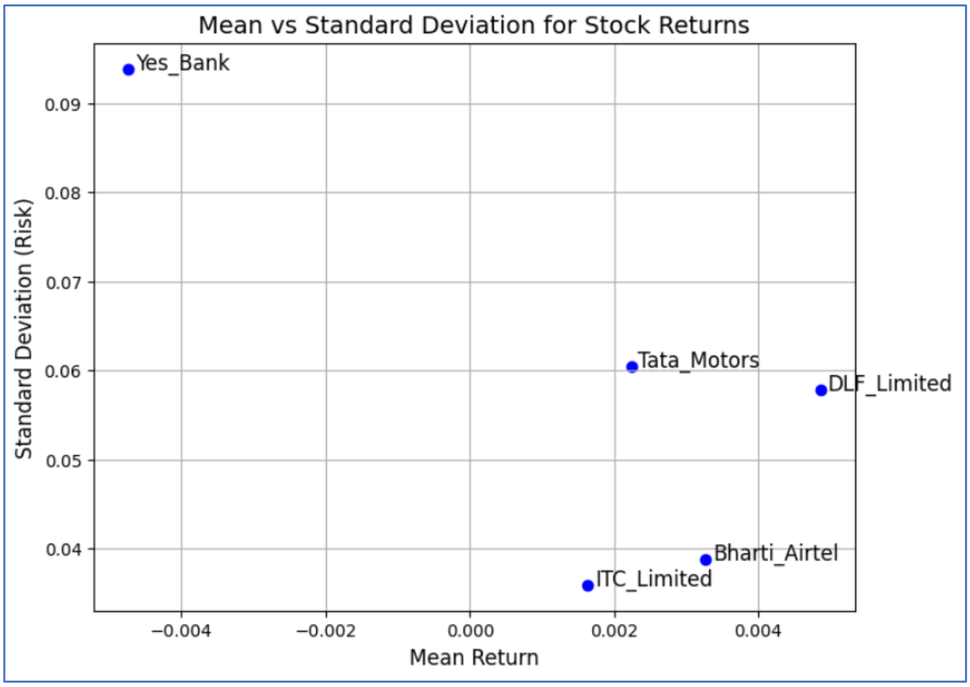

Fig 6. Mean vs Standard Deviation Plot

OBSERVATIONS AND INFERENCES:

A. ITC Limited

- The stock has the lowest average return among the five stocks, indicating that it offers relatively stable, but low growth in terms of price increase. Its volatility is also on the lower side, meaning it is less risky compared to the others.
- This stock is likely to be favoured by conservative investors seeking stability, as it provides steady returns with relatively low risk.

B. Bharti Airtel

- The stock has a slightly higher average return than ITC, suggesting that it provides better growth prospects. Its volatility is slightly higher than ITC's, indicating that it carries a bit more risk, but the return seems to justify it.
- This stock may appeal to investors looking for moderate growth with moderate risk.

C. Tata Motors

- The stock shows a modest average return, but it has relatively higher volatility compared to both ITC and Bharti Airtel.
- Investors looking for higher returns and who are willing to tolerate more risk might find Tata Motors attractive, but it could be considered less stable than others.

D. DLF Limited

- The stock shows the highest average return of the group, suggesting that it has the highest growth potential. The stock price is more unpredictable and has a higher risk.
- This might be appealing to risk-tolerant investors seeking high returns, but it requires the ability to handle significant price fluctuations.

E. Yes Bank

- The stock has negative average return which indicates that the stock has been underperforming over the period analysed. The stock is highly volatile and subject to large price fluctuations. This combination of negative returns and high volatility suggests that Yes Bank carries significant risks and has not performed well over the period.
- It may not be an attractive option for most investors unless there is a specific strategy in place for trading volatile stocks.

9. Actionable Insights & Recommendations

A. **ITC Limited** offers the most stability with low volatility and relatively low returns. It might be suitable for conservative investors. If low risk is the criteria, ITC Limited would be a good choice, but the return is modest.

B. **Bharti Airtel** offers a balance of reasonable returns with moderate risk, making it a good option for investors looking for safer, but relatively more rewarding opportunities. If moderate risk and moderate returns is the criteria, Bharti Airtel could be a solid pick.

C. **Tata Motors** and **DLF Limited** have higher returns, but they come with much higher volatility. These stocks are suitable for risk-tolerant investors willing to embrace price swings for higher returns. If high risk for potentially high returns is the criteria, Tata Motors and DLF Limited may be more suitable.

D. **Yes Bank** stands out as a high-risk, underperforming stock with negative returns. Its high volatility suggests it may not be suitable for most investors unless it shows signs of improvement or there is a specific market opportunity. It should likely be avoided unless you have a specific reason for considering its volatility or believe it has future recovery potential.